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الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM129
Technologies in Practice

Midterm Examination (MTA)
Summer Semester 2025/2026

Time Allowed: 2 hours

Instructions:

- Total Marks: **60**
- This exam consists of **4** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Question 1: Choose the correct answer [10 Marks]

1. Which component of an operating system manages the execution of processes?

a) File Manager
b) Process Manager
c) Compiler
d) Linker

2. Which OSI layer is responsible for logical addressing?

a) Physical Layer
b) Data Link Layer
c) Network Layer
d) Session Layer

3. Which protocol is used to dynamically assign IP addresses to devices on a network?

a) DNS
b) FTP
c) DHCP
d) HTTP

4. Which of the following is a valid IPv4 address?

a) 192.168.1.300
b) 10.0.0.1
c) 256.100.10.5
d) 172.16.-1.1

5. What is the default subnet mask of a Class C IPv4 address?

a) 255.0.0.0
b) 255.255.0.0
c) 255.255.255.0
d) 255.255.255.255

6. Which network device forwards packets between different networks?

a) Hub
b) Switch
c) Router
d) Repeater

7. What does DNS stand for?

a) Data Naming Service
b) Domain Name System
c) Dynamic Network Service
d) Digital Naming System

8. Which memory is volatile?

a) ROM
b) Hard Disk
c) SSD
d) RAM

9. Which command is used to display the current IP configuration in Windows?

a) ping
b) ipconfig

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c) netstat
d) tracert

10. Which OSI layer is responsible for end-to-end communication?

a) Transport Layer
b) Session Layer
c) Application Layer
d) Physical Layer

Question 2: Compare between the following [15 Marks]

a) Compare between DHCP and NAT? (3 Marks)
b) Compare Twisted-Pair cable and Fiber optic cable? (3 Marks)
c) Compare between User mode and Kernel mode? (3 Marks)
d) Compare between MAC address and IP address? (3 Marks)
e) Compare GUI and CLI Interfaces? (3 Marks)

Question 3: Answer the following questions [5 x 5 marks each = 25 Marks]

a) Briefly explain how multiprogramming increases CPU utilization?
b) Contrast the security capabilities of WPA2 and WEP protocols in securing wireless networks. What are the advantages and limitations of each?
c) Explain the purpose of port numbers in network communication and define the three categories of port numbers as assigned by ICANN.
d) Explain about the purpose and functions of linker and loader in an Operating system.
e) Match the networking concept of Column A with Column B

Column A	Column B
1. SaaS (Software as a Service)	**A.** A protocol used to transfer web pages from a server to a client browser.
2. HTTP	**B.** A network security device that monitors and controls incoming and outgoing network traffic based on security rules.
3. IPv6	**C.** A cloud service model that provides software applications over the internet on a subscription basis.
4. Twisted Pair Cable	**D.** A 128-bit Internet Protocol address designed to replace IPv4.
5. Firewall	**E.** A type of network cable consisting of pairs of insulated copper wires twisted together to reduce interference.

Question 4: Answer the following questions [20 Marks]

a) Explain about unicast transmission, broadcast transmission, and multicast transmission each with an example diagram. (9 Marks = 2 marks for each explanation + 1 marks for each diagram).

b) If a company is assigned the IP address 192.168.45.12. Answer the following questions (5 marks = 1 mark + 1 mark + 1 mark + 2 marks)

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1) Identify the class of this IP address.
2) What is the default subnet mask for this class?
3) Determine the network portion and host portion of the address.
4) Calculate the maximum number of hosts supported in this class by default.

c) A family has moved into a new house and wants to connect laptops, smartphones, a smart TV, and a printer to the Internet. They prefer a solution that allows mobility throughout the house and easy addition of new devices. (6 marks = 2 marks each)

a) Would you recommend a wired or wireless network? Justify your answer.
b) What is the purpose of an SSID in a wireless network?
c) State two advantages of using wireless networking in this scenario.

End of questions

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